CHAPTER 3: RESULTS AND DISCUSSION

1. Presentation of the Developed Project

Project Title

CogniCore: An Adaptive Cognitive Training Game with AI-Driven Skill Assessment and Analytics

Purpose

CogniCore is a web-based gamified cognitive training platform designed to replace traditional, anxiety-inducing neuropsychological assessments with an engaging and interactive alternative. Instead of formal clinical tests, users complete real-time mini-games that measure their cognitive performance across four domains — Reflexes & Focus, Spatial-Visual Memory, Logical-Mathematical Reasoning, and Executive Strategy — while the system automatically adapts the difficulty and analyzes their results using machine learning.

Target Users

BSIT and BSCS students aged 18–25 who seek a non-clinical, low-stress method of assessing and improving their cognitive capabilities as part of academic wellness and performance

Platform & Technology Stack

Layer	**Technology**	**Version**
Game Engine	Phaser.io	v3.80.1
Frontend / Dashboard	React.js	v18.3.1
Frontend Build Tool	Vite	v5.4.21

State Management	Zustand	v5.0.14
Charts & Analytics UI	Chart.js + react-chartjs-2	v4.5.1
Backend	Python / Flask	v3.0.0
ORM	Flask-SQLAlchemy	v3.1.1
Database	PostgreSQL (hosted via **Supabase**)	—
Authentication	PyJWT + bcrypt	v2.8.0 / v4.1.2
Machine Learning	Scikit-learn (Random Forest Classifier + KMeans Clustering + IsolationForest)	v1.9.0
Numerical Computing	NumPy + SciPy	v2.2.6 / v1.17.1
Offline Data Storage	IndexedDB (Browser API)	—
PWA Support	vite-plugin-pwa	v1.3.0

Machine Learning Integration: The system integrates a Scikit-learn Random Forest Classifier (cognitive_model.pkl) to classify players into cognitive archetypes (e.g., Fast Learner, Precision Expert, High Fatigue) based on their real-time session telemetry. It also uses KMeans Clustering (clustering_model.pkl) for unsupervised grouping and an IsolationForest for anomaly/fatigue detection.

Minimum System Requirements (to run locally)

Requirement	Minimum Specification
Operating System	Windows 10 / macOS 12 / Ubuntu 20.04
Node.js	v18.0.0 or higher

Python	v3.12
pip	v23.0+
RAM	4 GB minimum (8 GB recommended)
Browser	Google Chrome / Firefox (latest version) with WebGL support (Desktop versions only; mobile browsers are not supported).
Internet Connection	Required for database sync (Supabase is cloud-hosted)
GPU	Any GPU with WebGL support (for Phaser rendering)

To run the project locally:

1. Install Node.js dependencies → npm install (inside /client)
2. Install Python dependencies → pip install -r requirements.txt (inside /server)
3. Start backend → python app.py
4. Start frontend → npm run dev

2. Project Description

Major Modules — The Four Cognitive Domains & Their Mini-Games

CogniCore is organized into **four cognitive training domains**, each containing multiple mini-games built using the Phaser.io game engine:

Domain 1: Reflexes & Attentional Focus

Trains reaction speed, sustained attention, and resistance to visual interference.

Game	What Changes Per Difficulty Level
Speed Tap	Spawn delay, target lifespan, target size, distractor count
Focus Finder	Distractor count(5 → 25), distractors start moving at Level 3
Memory Match	Grid size, sequence length, flash duration

Stroop Shift	Spawn delay, conflict probability, text starts spinning at Level 5
Mental Flex	Rule shift frequency, decision time, number of rule types
Equation Balance	Number range, operators expand to include, time limit
Sequence Decoder	Sequence length, pattern types expand to Fibonacci/dual-rule, time
Route Optimizer	Node count (4→8 nodes), weight range, time limit

The DDA system collects performance telemetry — reaction time (in milliseconds), accuracy rate, error count, and hesitation patterns — after every trial. The server evaluates this data and returns updated difficulty parameters before the next round begins. This keeps the player in the optimal learning zone: not too easy to be boring, not too hard to be frustrating.

2. Distractor Management

As requested by Sir Rob, CogniCore implements multi-layer Distractor Management that activates progressively as difficulty increases:

Visual Distractors:

- **Focus Finder** → distractor count grows from 5 to 25 objects, and they begin moving at Level 3+
- **Speed Tap** → distractor objects appear at up to 40% spawn probability at Level 5
- **Logic Link** → distractor answer choices increase from 0 to 4 per question
- Stroop Shift → at Level 4, stimulus text rotates statically; at Level 5, it spins dynamically and random flash distractors appear on screen
- **Priority Queue** → category labels are progressively hidden (ambiguity increases), making task sorting harder

Auditory Distractors (via the CogniFX Audio Engine):

- At **difficulty Level 4 and above,** a background noise oscillator activates — a low-frequency square wave hum with LFO (Low-Frequency Oscillation) modulation that simulates real-world interference (like background office noise or a crowded cafeteria)
- At **Level 5,** the noise intensity doubles to push the player's focus to its limit
- Players with accessibility needs can **disable auditory distractors** from the Settings panel without affecting their score

Workflow — The Complete User Journey

1. LOGIN

- User creates an account (username + PIN) or logs into existing account.
- Authentication is handled via JWT (JSON Web Token) secured with bcrypt encryption.

2. DASHBOARD

- User sees their Cognitive Profile, domain scores, XP, coins, daily tasks,
- achievement badges, and AI-generated Pros & Cons of their cognitive strengths and weaknesses.

3. SELECT A GAME

- The user picks any of the 14 mini-games from the dashboard.
- They choose a Game Mode (Standard Timed, Zen, Survival, Time Attack,
- Objective Target, or Fatigue Endurance) from the Game Launcher modal.

4. PLAY

The Phaser.js game engine launches the selected game.
Every player action generates telemetry:

- ❖ Reaction time (milliseconds per trial)
- ❖ Accuracy (correct / incorrect)
- ❖ Hesitation patterns (time between stimulus and first input)
- ❖ Error counts and spam click detection

5. REAL-TIME DDA

- After every trial, telemetry is sent to the Flask backend via API.
- The server runs the Adaptive Task Staircasing algorithm and returns
- new difficulty parameters for the next round. This loop continues
- throughout the entire game session.

6. SESSION END

Final session stats (score, accuracy, average reaction time, difficulty reached) are submitted to the PostgreSQL/Supabase database.

7. AI ANALYSIS

The backend's Random Forest Classifier analyzes the session data:

- Reads the player's reaction time trend (using OLS Linear Regression slope)
- Reads accuracy rate and consistency
- Classifies the player into one of three Cognitive Archetypes:

- "Fast Learner" — High speed, consistent improvement
- "Precision Expert" — High accuracy, methodical approach
- "High Fatigue" — Declining performance over time, inconsistent

8. DASHBOARD UPDATE

The user returns to the dashboard and sees:

- Their updated Cognitive Archetype label
- Domain-specific scores updated across all 4 domains
- AI-generated Pros & Cons text (e.g., "Your Visual Memory is excellent!")
- XP gained, coins earned, achievements unlocked
- Performance charts and trends over time

9. RESEARCHER PORTAL (Admin Only)

Researchers/professors can log into the Admin Panel to view:

- All student session data in aggregate
- Comparative analytics across the entire cohort
- Individual player performance trends

3. Project Structure

A. DIAGRAMS:

1. The Site Map

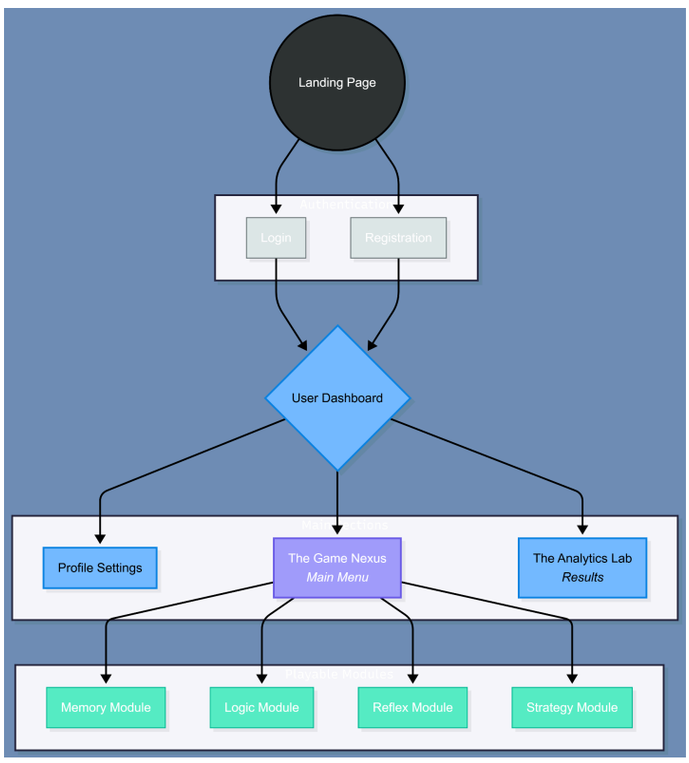


Figure 1. CogniCore Site Map and Navigation Architecture

This figure illustrates the hierarchical navigation structure of the CogniCore web application. It maps the user journey from the initial landing and authentication portal, through the central Game Nexus, and branching into the specialized cognitive training modules and the AI Analytics Lab. This ensures a seamless and intuitive user experience.

2. The System Architecture

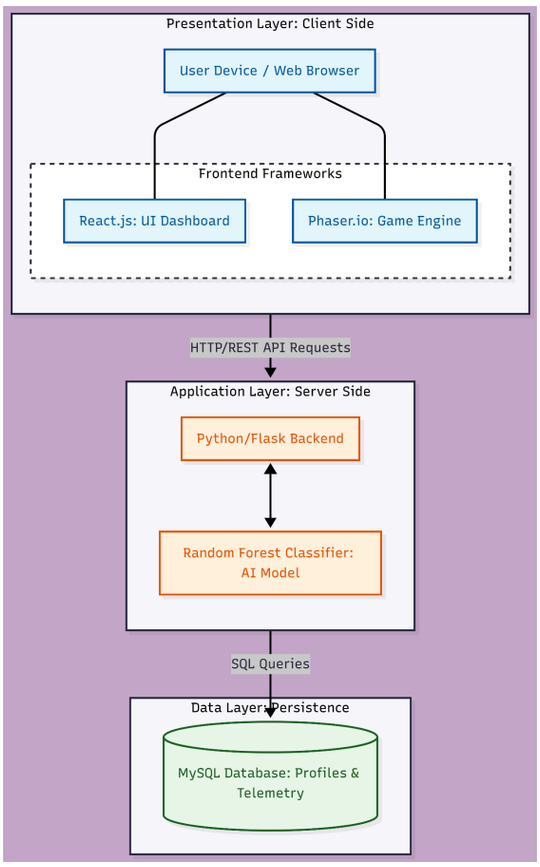


Figure 2. CogniCore System Architecture

This diagram presents the high-level system architecture of CogniCore, detailing the interaction between the client-side presentation layer (React.js and Phaser.io), the application server (Python/Flask API), and the data storage layer (MySQL). It visually demonstrates how gameplay telemetry is securely transmitted to the server for processing by the Random Forest Classifier.

3. The Data Flow Diagram

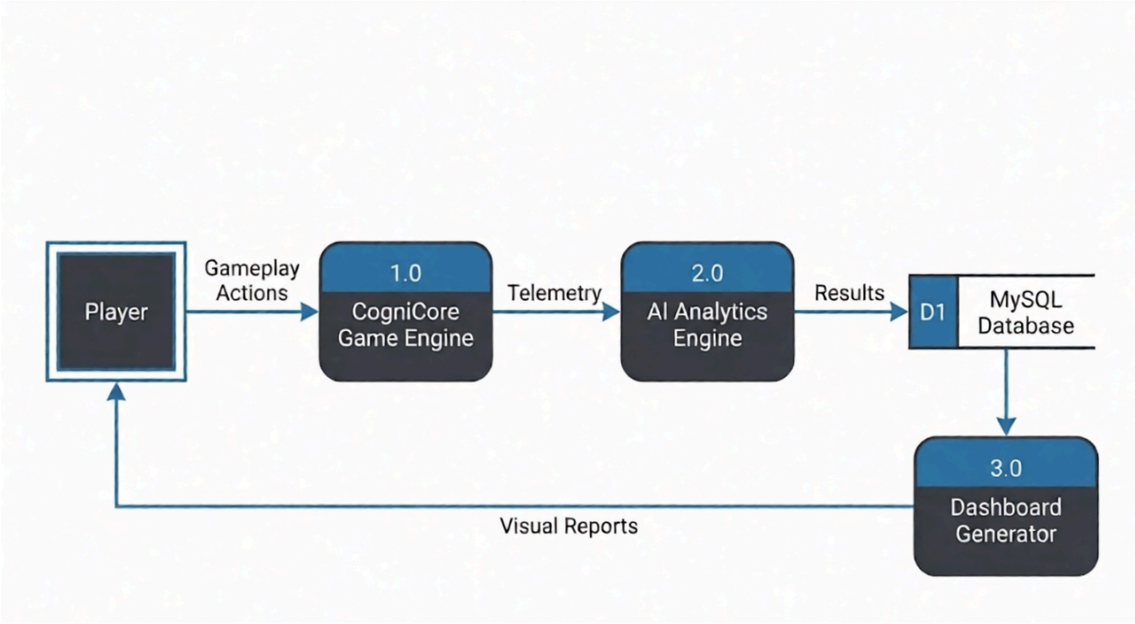


Figure 3. Level 0 Data Flow Diagram (Gane-Sarson Notation)

Utilizing Gane-Sarson notation, this figure demonstrates the primary flow of information within the system. It outlines how external player inputs and raw gameplay telemetry are captured by the game engine (Process 1.0), analyzed by the Machine Learning model (Process 2.0), securely stored in the MySQL database (Data Store D1), and ultimately retrieved to generate visual cognitive reports for the user via the dashboard (Process 3.0).

4. The Program Flowchart

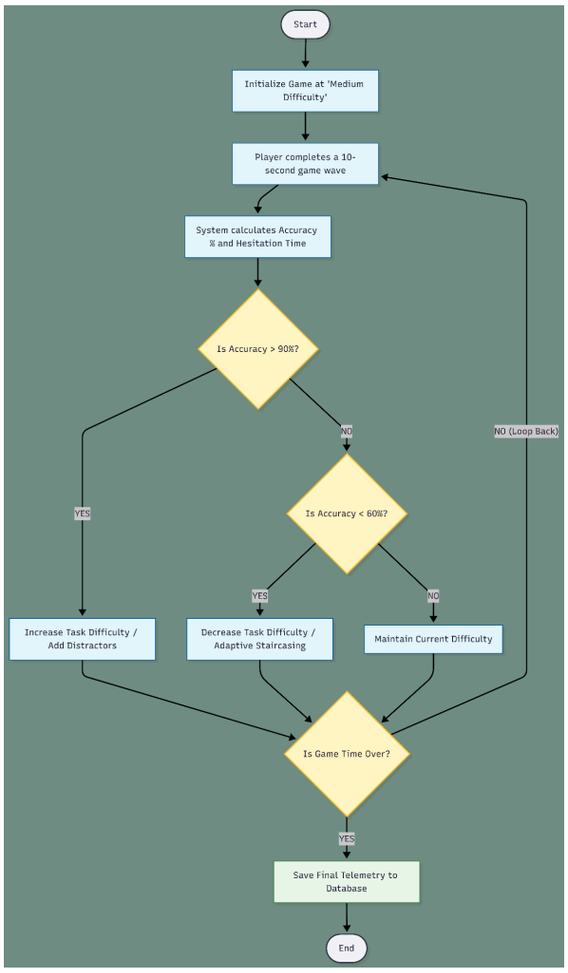


Figure 4. Dynamic Difficulty Adjustment (DDA) Program Logic Flowchart

This flowchart maps the logical decision-making process of the system's Dynamic Difficulty Adjustment (DDA) feature. It details the programmatic loop where the system evaluates real-time player accuracy and hesitation time, subsequently deciding whether to increase difficulty, decrease difficulty via adaptive staircasing, or introduce visual distractors.

B. GUI Interface

1. LOGIN PAGE

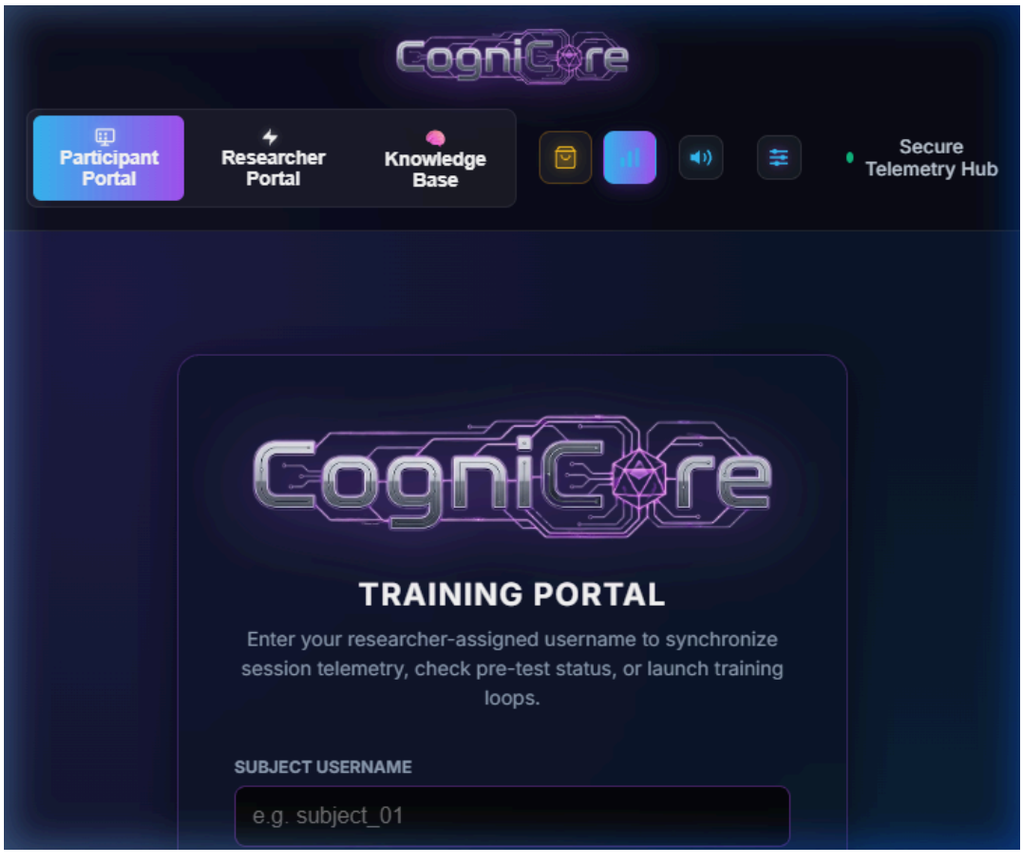


Figure 1 — CogniCore Login Portal

The CogniCore login portal presents users with two entry points: a Participant Portal for student users and a Researcher Portal for faculty and administrators. Students authenticate by entering their registered username and agreeing to a data collection consent disclaimer before proceeding to their personalized dashboard.

2. Main Dashboard



Figure 2 — User Dashboard (Cognitive Profile Overview)

The User Dashboard serves as the central hub of the CogniCore platform, displaying the student's AI-assigned Cognitive Archetype, domain performance scores across four cognitive categories, XP level, coin balance, daily training tasks, and streak information. The dashboard aggregates all historical session data to provide the user with a real-time snapshot of their cognitive training progress.

3. **Game Hub**

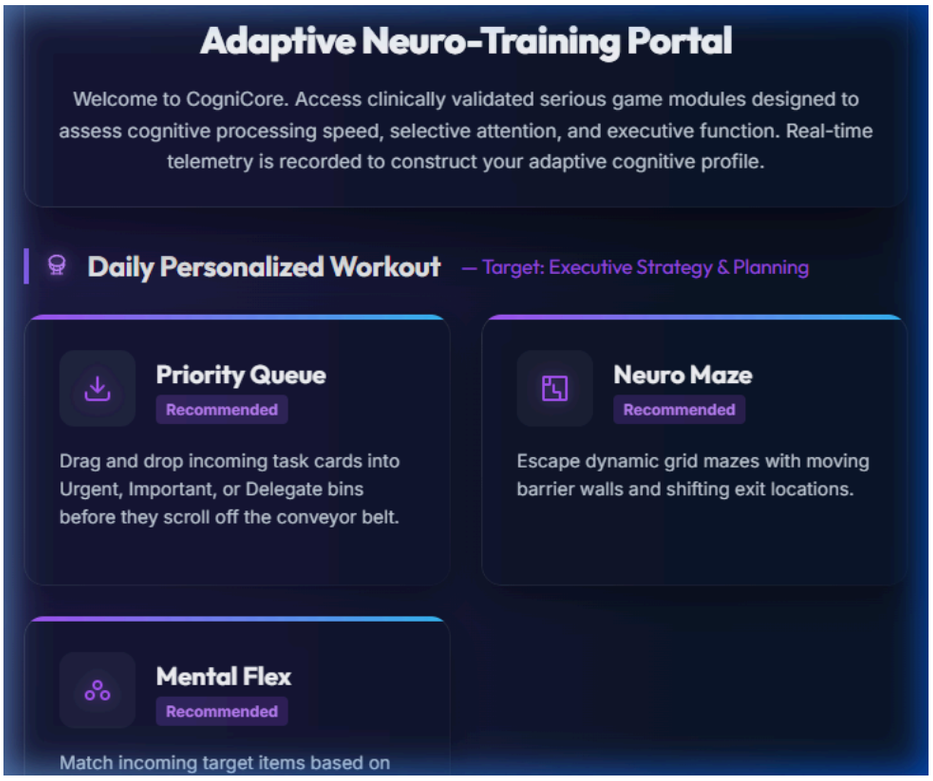


Figure 3 — Game Hub (Training Module Selection)

The Game Hub presents all available cognitive training mini-games organized by their respective cognitive domains — Reflexes & Attentional Focus, Spatial-Visual Memory, Logical-Mathematical Reasoning, and Executive Strategy. Each game card displays the game title, a brief description, and a personalized training recommendation button generated by the AI system.

4. **Game Launcher**

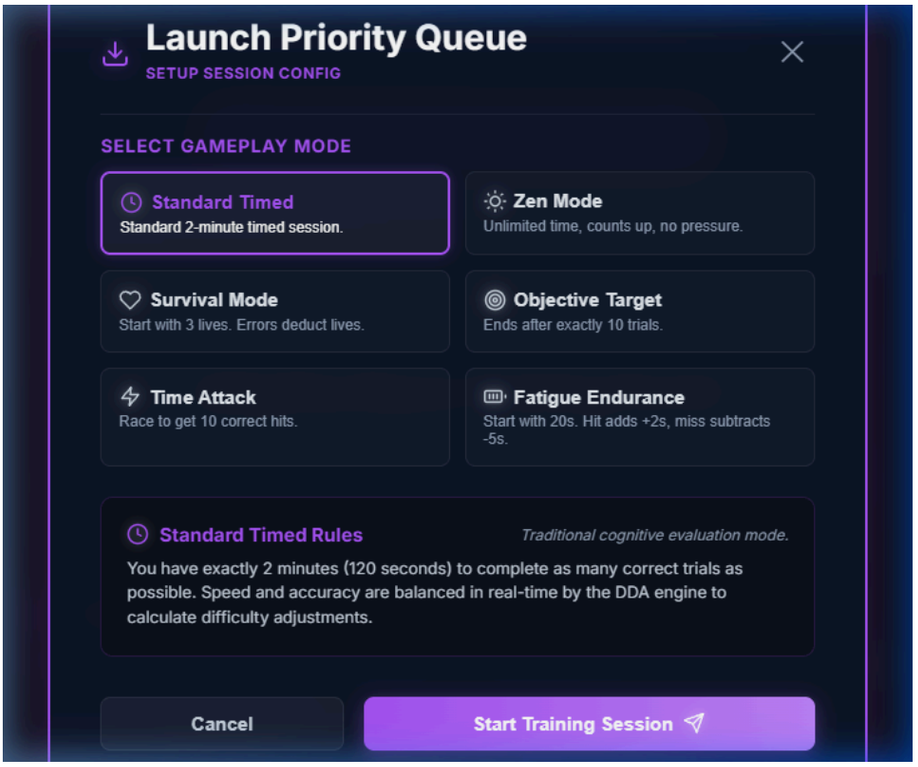


Figure 4 — Game Launcher Modal (Session Configuration)

The Game Launcher Modal allows users to configure their training session before beginning a game by selecting from six available gameplay modes: Standard Timed, Zen Mode, Survival Mode, Objective Target, Time Attack, and Fatigue Endurance. Each mode modifies the session's scoring conditions and timer behavior, giving users control over the cognitive challenge format that best suits their training objectives.

5. **Research Portal**

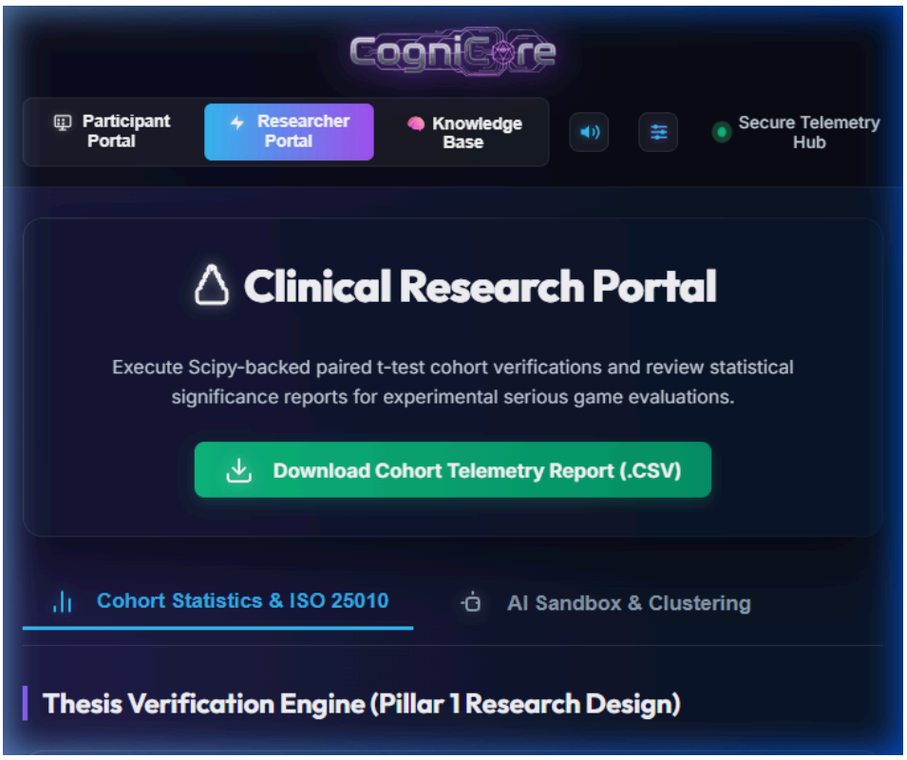


Figure 5 — Researcher Portal (Analytics Dashboard)

The Researcher Portal provides authorized faculty members and administrators with a comprehensive view of aggregated student performance data across all cognitive domains and game sessions. It presents cohort-level analytics, individual session breakdowns, and AI-generated performance trends to support academic research and student cognitive wellness monitoring.

6. Overview Tab (User Profile)

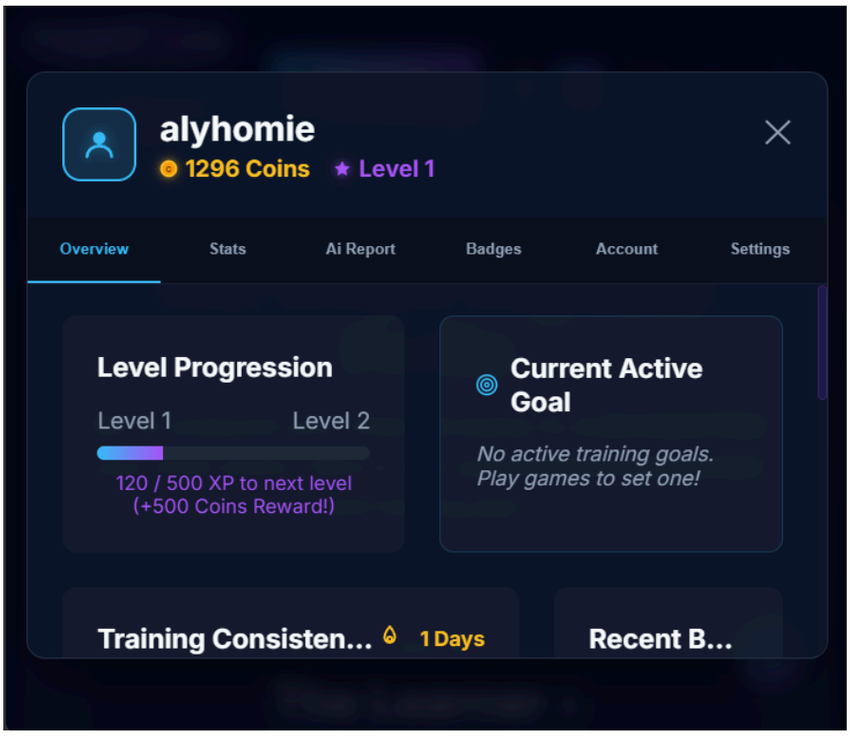


Figure 6 — User Profile Overview

The Overview Tab acts as the player's personal summary page. It provides an immediate, at-a-glance view of their current trajectory. It features the player's "Current Active Goal" with a progress bar, a "Training Consistency" heat map (showing their activity over the last 21 days), and a "Recent Activity" log detailing their latest gameplay sessions. This interface is designed to keep users motivated by visualizing their daily commitment and short-term milestones.

7. Stats Tab (Performance Analytics)



Figure 7 — Detailed Cognitive Statistics

The Stats Tab provides a deeper, quantitative breakdown of the user's cognitive performance. It utilizes a Radar Chart to visualize the balance between the four core cognitive domains (Reflexes, Memory, Logic, Strategy). Additionally, it provides historical line graphs tracking accuracy and reaction times over multiple sessions. This interface allows users to pinpoint exactly which cognitive skills are improving and which require more focused training.

8. AI Report Tab (Machine Learning Insights)

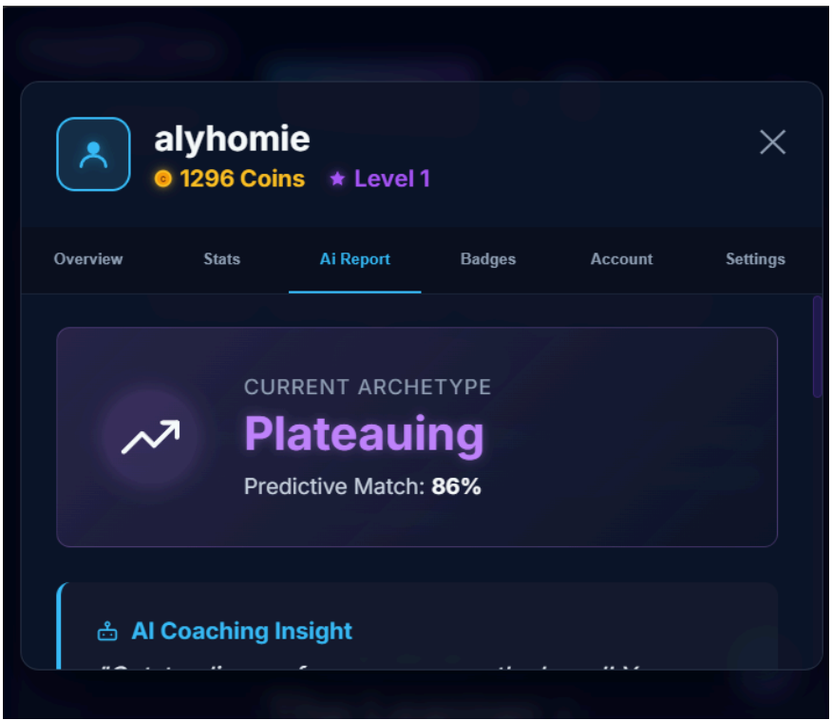


Figure 8 — AI-Generated Cognitive Report

The AI Report interface is where the system's Machine Learning backend directly communicates with the user. It displays the user's classified "Cognitive Archetype" (e.g., Fast Learner, Plateauing) along with a confidence percentage. Below this, a personalized "AI Coaching Insight" provides dynamically generated, qualitative feedback based on recent gameplay telemetry. It also highlights the user's Top Strength and Primary Bottleneck, recommending specific games to address their weaknesses.

9. Badges Tab (Gamification & Achievements)

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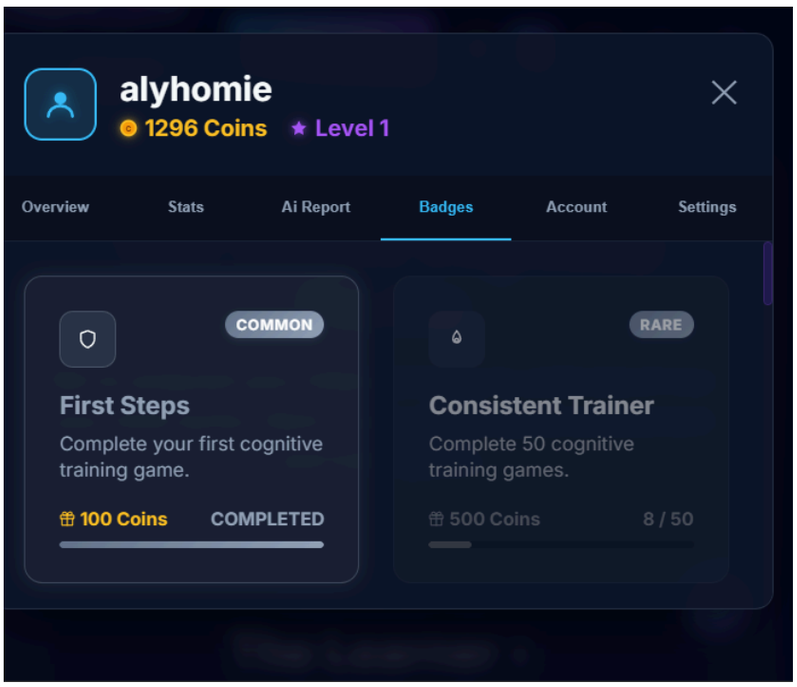


Figure 9 — Achievements and Badges Interface

The Badges Tab is a core component of the platform's gamification strategy. It displays a gallery of unlockable achievements, categorized by rarity (Common, Rare, Epic, Legendary). Each badge card shows the specific requirement, the user's current progress toward unlocking it (via a progress bar), and the coin reward upon completion. This interface leverages completionist psychology to encourage long-term user retention.

10. Gameplay Interface (Mini-Game Example)



Figure 10 — Standard Gameplay Interface (e.g., Speed Tap)

The Gameplay Interface is rendered using the Phaser.js engine and is designed to be distraction-free to ensure accurate cognitive measurement. The UI displays critical real-time data at the top of the screen, including the current Score, remaining Time, Multiplier, and Lives (depending on the selected game mode). The central canvas is reserved entirely for the cognitive stimuli, ensuring the user's focus remains unbroken during the assessment.

11. Profile Modal: Account & Settings Tab

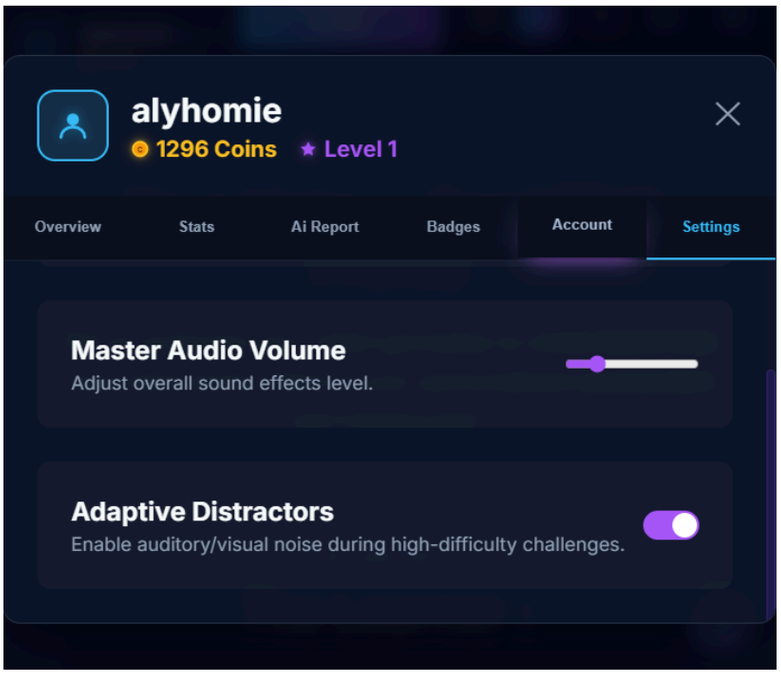


Figure 11 — Account and Settings Configuration

The Settings interface provides users with control over their application experience. This includes audio configuration (master volume, SFX, background music) and critical accessibility toggles, such as disabling flashing visual effects or auditory distractors. It also houses account management functions like changing the display name, resetting passwords, and viewing the terms of service.

12. Seizure Disclaimer & Accessibility Prompt

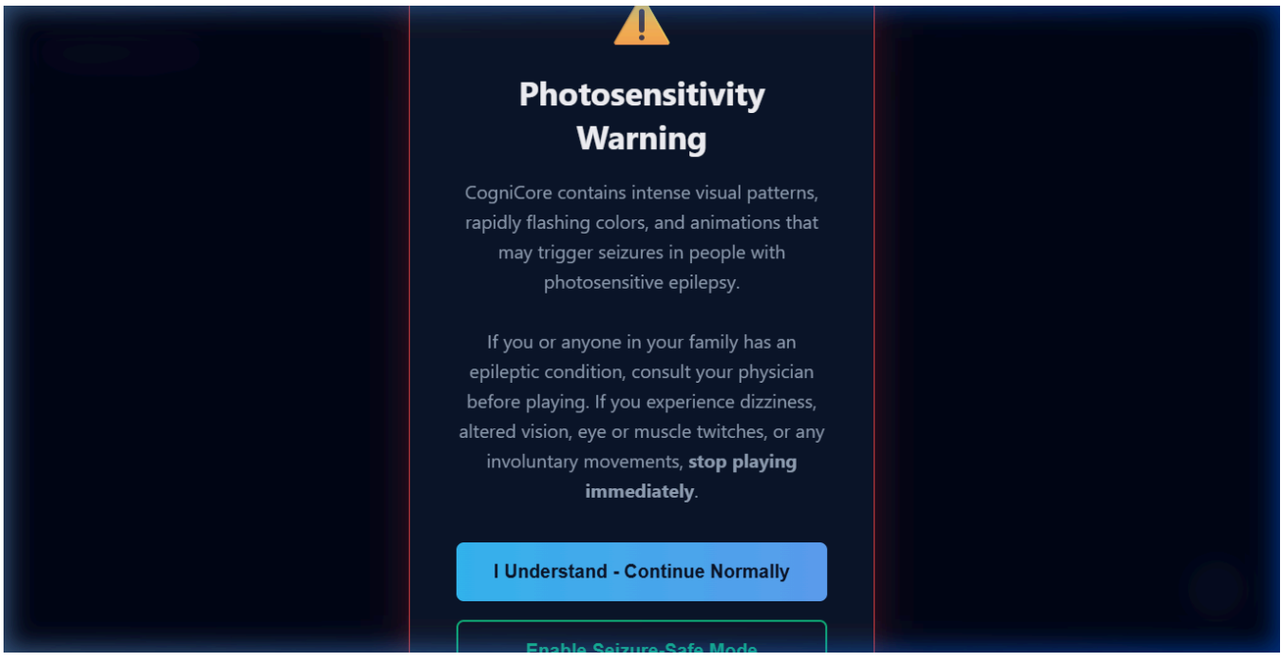


Figure 12 — Health Disclaimer and Accessibility Prompt

To adhere to software safety standards, this interface is presented to the user to warn them about potential photosensitive triggers (such as rapidly flashing elements in higher difficulties). It requires the user to acknowledge the risk before proceeding and provides an immediate toggle to enable Accessibility Mode, which turns off intense visual strobing.

13. Pre-Test / Post-Test Results Screen



Figure 13 — Session Results and Performance Summary

Immediately following a gameplay session, the user is presented with the Results Interface. This screen provides an immediate feedback loop, displaying the final score, accuracy percentage, average reaction time, and the amount of XP and Coins earned. It serves to reinforce learning and provides instant gratification for completing a cognitive task.

14. The Reward Shop

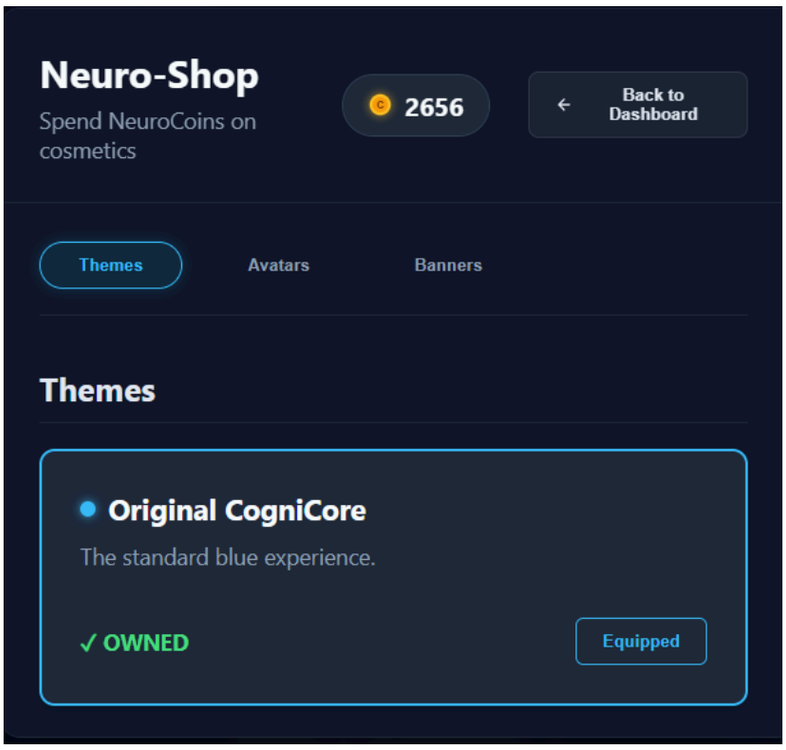


Figure 14 — The Virtual Reward Shop

The Shop interface is a key gamification element that gives the in-game currency ("Coins") tangible value. Users can spend coins earned from training sessions to purchase cosmetic upgrades, profile banners, or avatar frames. This GUI mimics standard video game economies, transforming a clinical cognitive assessment into an engaging, replayable experience.

15. Global Leaderboards



Figure 15 — Competitive Leaderboard Interface

The Leaderboard interface introduces a social and competitive dimension to the platform. It ranks users based on their total XP, highest scores in specific mini-games, or longest training streaks. This interface appeals to players motivated by social comparison, driving them to consistently return and improve their cognitive metrics to climb the ranks.

16. In-Game Pause Menu & Live HUD

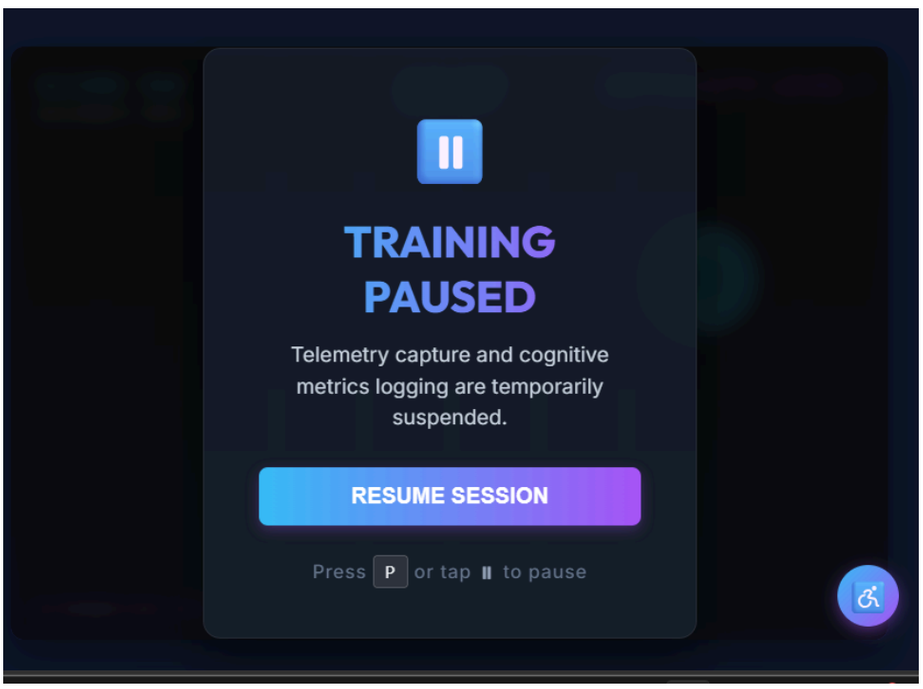


Figure 16 — Pause Overlay and Live Telemetry HUD

The In-Game Pause Menu provides standard game controls (Resume, Restart, Quit) while the Live HUD (Heads-Up Display) can optionally show real-time Dynamic Difficulty Adjustment (DDA) metrics. If enabled, the HUD visualizes the background algorithm at work, showing the user how their accuracy and reaction time are actively manipulating the game's difficulty parameters in real time.

17. Knowledge Base (Help Center)

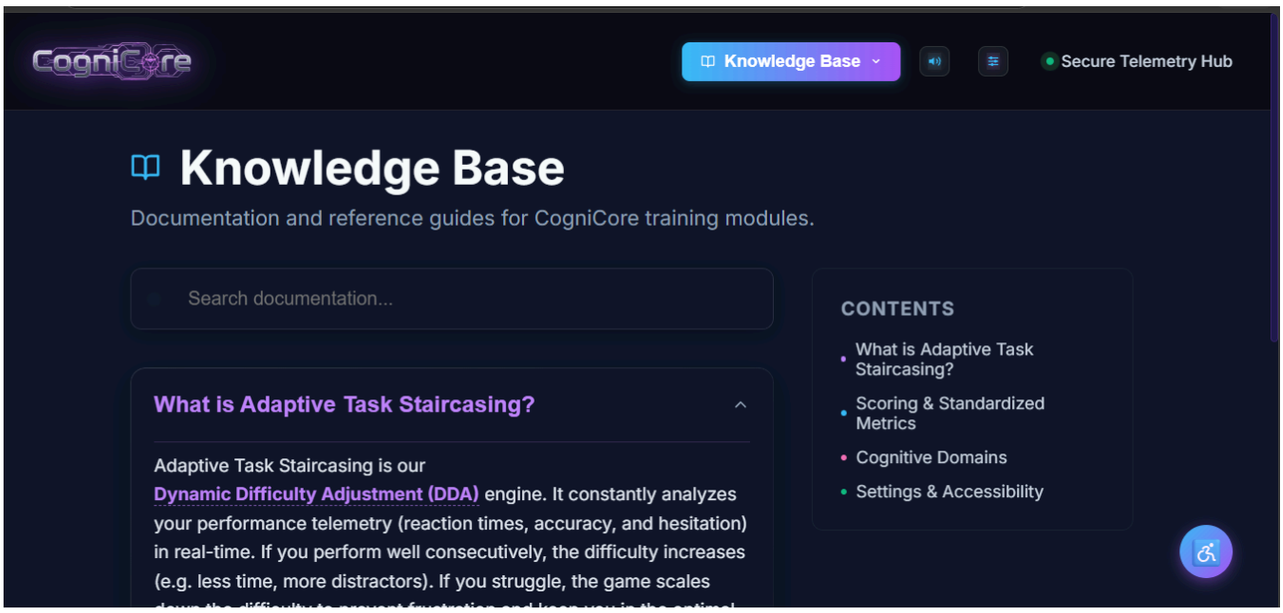


Figure 17 — Interactive Knowledge Base

The Knowledge Base interface serves as the educational backbone of CogniCore. It provides users with accessible, easily digestible information about cognitive science, explaining the real-world applications of the skills they are training (e.g., Executive Function, Spatial Memory) and providing answers to frequently asked questions regarding how the AI evaluates them.

4. System Test Plan

Testing Matrix:

Module 1: Authentication & Login System

#	Test Case	Objective	Expected Result	Actual Result	Status

TC-01	Enter a valid registered username and correct PIN	Verify successful login	User is authenticated, JWT token is issued, and redirected to the Dashboard	User was authenticated and Dashboard loaded successfully	Pass
TC-02	Enter a valid username with an **incorrect PIN**	Verify login rejection on wrong credentials	System displays an error message and denies access	Error message displayed; login was rejected	Pass
TC-03	Leave the username field **blank** and attempt login	Verify input validation for empty fields	System prevents submission and shows a validation warning	Form submission was blocked; warning message appeared	Pass
TC-04	Register a **new account** with a unique username and 4-digit PIN	Verify account creation and database write	New user record is saved in the `users` table in Supabase; user is redirected to Dashboard	New account was created successfully; Dashboard loaded	Pass
TC-05	Attempt to register a username that **already exists**	Verify duplicate username prevention	System displays "Username already taken" and blocks registration	Duplicate detected; registration was blocked	Pass
TC-06	Log in as a **Researcher** using the Researcher Portal entry point	Verify role-based access control	Researcher is redirected to the Admin/Researcher Panel instead of the student Dashboard	Researcher Portal loaded with cohort analytics view	Pass
TC-07	Close and reopen the browser tab **without logging out**	Verify session persistence via JWT	User session is restored automatically without re-entering credentials	Session was restored via stored token	Pass

Module 2: Database Saving & Data Integrity

#	Test Case	Objective	Expected Result	Actual Result	Status
TC-08	Complete a full game session (Speed Tap, Standard Timed mode)	Verify game session is saved to the database	A new record is created in `game_sessions` and all trial data is written to `performance_metrics` in Supabase	Session and all trial metrics were recorded correctly	Pass
TC-09	Complete a game session and check the **Dashboard** for updated scores	Verify domain score recalculation after session	The Reflexes & Focus domain score on the Dashboard updates to reflect the new session	Domain score updated correctly after session	Pass
TC-10	Play a game while **disconnecting from the internet** mid-session	Verify offline data buffering via IndexedDB	Trial telemetry is saved locally in the browser's IndexedDB; data is synced to the server when connection is restored	Data was buffered offline; successfully synced upon reconnection	Pass
TC-11	Complete 3 game sessions in one day and check **Daily Tasks**	Verify daily task progress tracking	The "Play 3 Training Games" daily task counter increments to 3 and marks as complete	Daily task progress updated correctly to 3/3	Pass
TC-12	Achieve over 80% accuracy in any game and check **Daily Tasks**	Verify accuracy-based task tracking	The "Achieve 80% accuracy" daily task is marked complete	Task correctly marked as complete after high-accuracy session	Pass

TC-13	Complete a session and verify **XP and Coins** are awarded	Verify gamification reward calculation	XP = difficulty × 15; Coins = (accuracy × 10) + (difficulty × 2) are correctly added to user profile	XP and Coins were accurately calculated and added	Pass
TC-14	Accumulate enough XP to **level up**	Verify level progression logic	When XP exceeds the level threshold (500 XP per level), level increments and 500 bonus coins are awarded	Level-up triggered correctly with bonus coins	Pass

Module 3: Game Stability & Crash Testing

#	Test Case	Objective	Expected Result	Actual Result	Status
TC-15	Launch all **14 mini-games** one by one	Verify all games load without errors	All 14 games initialize and render the Phaser canvas without JavaScript errors	All 14 games loaded successfully without crashes	Pass
TC-16	Launch a game and **immediately close/navigate away** from the browser tab	Verify no memory leaks or zombie processes	Game session is cleanly terminated; no errors in browser console	Game cleaned up properly upon tab close	Pass

TC-17	Play **Speed Tap** and intentionally click only **distractor objects** (wrong targets) 5 times in a row	Verify miss handling and survival mode life deduction	In Survival Mode: lives decrement from 3 → 0 and game ends. In Standard mode: miss counter increments correctly	Lives decremented correctly; game ended at 0 lives	Pass
TC-18	Play **Memory Match** and intentionally recall the sequence **incorrectly** every round	Verify incorrect input handling	Error count increments; DDA evaluates performance and lowers difficulty in next round	Errors tracked; difficulty lowered in the subsequent round	Pass
TC-19	Launch **Synapse Spin** on a device without WebGL support	Verify graceful rendering fallback	Game falls back to Canvas renderer without crashing	Phaser's automatic renderer fallback activated; game still ran	Pass
TC-20	Play a game in **Zen Mode** for an extended period (5+ minutes)	Verify timer counts up indefinitely without crashing	Timer counts upward from 00:00 with no timeout or forced end	Timer ran correctly for extended duration; no crash	Pass
TC-21	Play **Fatigue Endurance** mode and intentionally miss repeatedly	Verify time penalty logic	Each miss subtracts 5 seconds from the timer; game ends when timer reaches 0	"-5s" penalty applied correctly; game ended at 0 seconds	Pass
TC-22	Play **Time Attack** mode and get 10 correct hits	Verify win condition detection	Game ends immediately after the 10th correct hit and displays final results	Win condition triggered correctly at exactly 10 correct hits	Pass

TC-23	Resize the browser window during an active game session	Verify responsive canvas scaling	Phaser canvas scales proportionally without distorting game elements	Canvas resized correctly; no visual glitches observed	Pass
TC-24	Rapidly click the game start button **multiple times** in quick succession	Verify duplicate session prevention	Only one game session is initiated; no duplicate records in `game_sessions`	Single session launched; no duplicate database records	Pass

Module 4: Dynamic Difficulty Adjustment (DDA)

#	Test Case	Objective	Expected Result	Actual Result	Status
TC-25	Answer 5 consecutive questions incorrectly in Logic Link	Verify DDA lowers difficulty	DDA evaluates low accuracy, decrements difficulty level, and returns easier game parameters	Difficulty decreased from Level 3 to Level 2 after 5 errors	Pass
TC-26	Achieve 5 consecutive correct answers in Speed Tap with fast reaction times	Verify DDA raises difficulty	DDA evaluates high accuracy and speed, increments difficulty level, and returns harder game parameters	Difficulty increased; spawn delay decreased and distractor ratio increased	Pass
TC-27	Play Focus Finder at difficulty Level 5	Verify maximum distractor count	25 distractor objects appear on screen, all moving, with high visual similarity to the target	25 distractors appeared with movement enabled; similarity set to "high"	Pass

TC-28	Play Stroop Shift at difficulty Level 5	Verify all distractor types are active	Conflict probability = 100%, text rotates statically AND spins dynamically, and distractor flashes appear	All three distractor layers (rotation, spin, flashes) were active	Pass
TC-29	Play Mental Flex with known slow reaction times	Verify personalized time limit scaling	Time limit adjusts to: `(player's average RT × 1.2) + grace period` instead of the static default	Time limit dynamically adjusted to match player's personal reaction time baseline	Pass
TC-30	Disconnect from the internet during a DDA evaluation request	Verify DDA offline fallback	System returns a local fallback response (no difficulty change) and gameplay continues uninterrupted	DDA fallback triggered; game continued without crashing	Pass

Module 5: AI Archetype Classification

#	Test Case	Objective	Expected Result	Actual Result	Status
TC-31	Complete multiple sessions with consistently fast reaction times and high accuracy	Verify "Fast Learner" archetype classification	Random Forest Classifier predicts "Fast Learner" archetype based on speed and accuracy telemetry	"Fast Learner" archetype was correctly assigned on the Dashboard	Pass
TC-32	Complete multiple sessions with slow but highly accurate responses	Verify "Precision Expert" archetype classification	Classifier predicts "Precision Expert" based on high accuracy despite slower reaction times	"Precision Expert" archetype was correctly assigned	Pass

TC-33	Simulate declining performance across a session (good start, errors increase over time)	Verify "High Fatigue" archetype detection	IsolationForest anomaly detector identifies declining performance trend; classifier assigns "High Fatigue"	"High Fatigue" archetype was correctly detected and assigned	Pass
TC-34	Check the Pros & Cons section on the Dashboard after classification	Verify AI-generated text output	Dashboard displays domain-specific strength and weakness descriptions generated by the `generate_pros_cons()` function	Accurate, personalized Pros & Cons text appeared on the Dashboard	Pass
TC-35	Play exclusively games in the Logical-Mathematical domain	Verify domain-specific score update	Only the Logical-Mathematical domain score updates; other domain scores remain unchanged	Domain score updated selectively; other domains were unaffected	Pass
TC-36	Run the AI model when Scikit-learn is not installed	Verify deterministic rule-based fallback	System falls back to the built-in rule-based classifier; no crash occurs	Rule-based fallback activated; archetype classification still worked	Pass

Module 6: Gamification & Achievements

#	Test Case	Objective	Expected Result	Actual Result	Status
TC-37	Log in on **consecutive days**	Verify daily login streak tracking	Streak counter increments each day; daily reward popup appears with coin reward	Streak tracked correctly; reward popup appeared on login	Pass

TC-38	Complete an achievement condition (e.g., first game played)	Verify achievement unlock and notification	Achievement is saved to `user_achievements` table; an in-game notification popup appears	Achievement unlocked and notification displayed correctly	Pass
TC-39	Unlock a **score multiplier** as a "Fast Learner" archetype during gameplay	Verify archetype-based score bonus	Each correct hit adds a 50% speed bonus with floating "+X SPEED BONUS!" text on screen	Bonus calculated correctly at 1.5× multiplier; floating text appeared	Pass

Module 7: Researcher Portal & Analytics

#	Test Case	Objective	Expected Result	Actual Result	Status
TC-40	Log in as a Researcher and view the **Analytics Dashboard**	Verify researcher-only data access	Researcher Portal loads with aggregated cohort performance charts and individual student session data	Analytics dashboard loaded with all charts rendered correctly	Pass
TC-41	A student completes a session; verify it appears in the **Researcher Portal** in real time	Verify data propagation from game to analytics	New session data appears in the researcher's view after the student submits their results	Session data appeared in the portal immediately after submission	Pass
TC-42	Access the Researcher Portal using a **student account**	Verify role-based access restriction	Student is denied access to the Researcher Portal; an unauthorized error is returned	Access correctly denied; student could not view researcher data	Pass